

Python Interview Questions

1. What is Python?

Answer:

Python is a high-level, interpreted programming language used for web development, automation, data science, artificial intelligence, and machine learning.

Real-Time Example:

A company receives 5,000 customer emails every day. A Python program can automatically categorize and reply to them, saving time.

2. What are the Features of Python?

Answer:

- Easy to learn
- Interpreted language
- Object-oriented
- Platform-independent
- Open source
- Large standard library
- Dynamically typed

Real-Time Example:

The same Python application can run on Windows, Linux, and macOS without changing the code.

3. What is the Difference Between a List and a Tuple?

List

A list is an ordered, mutable collection of items in Python.

Mutable means you can add, remove, or change items after creating it.

Syntax: `my_list = [1, 2, 3, "apple"]`

A tuple is an ordered, immutable collection of items in Python.

Immutable means you cannot change it once created.

Syntax: `my_tuple = (1, 2, 3, "apple")`

Real-Time Example:

- List: Shopping cart items (items can be added or removed).
- Tuple: Date of Birth (cannot be changed).

4. What is a Variable?

Answer:

A variable is a named memory location used to store data.

Python Example:

```
name = "Bhavani"
```

```
age = 20
```

```
print(name)
```

```
print(age)
```

Real-Time Example:

When filling an online college application, the student's name, age, and email are stored in variables.

5. What are Python Data Types?

Answer:

Python supports different data types to store different kinds of values.

Python Example:

```
age = 20          # int
```

```
height = 5.4      # float
```

```
name = "Bhavani"  # string
```

```
passed = True     # boolean
```

Real-Time Example:

In a student management system:

- Roll Number → Integer

- Name → String
- Percentage → Float
- Fee Paid → Boolean

6. What is the Difference Between = and ==?

Answer:

- = → Assigns a value.
- == → Compares two values.

Real-Time Example:

- = → Saving a student's marks.
- == → Checking whether the student scored exactly 80 marks.

7. What is a Function?

Answer:

A function is a reusable block of code that performs a specific task.

Real-Time Example:

The Check Balance option in an ATM is a function. Whenever a customer selects it, the same function executes.

8. What is the Difference Between break and continue?

Break:A statement used to exit a loop immediately. When Python hits break, it stops the loop completely and moves to the code after the loop.

Continue:A statement used to skip the rest of the current loop iteration. When Python hits continue, it jumps straight to the next iteration of the loop.

Real-Time Example:

- break: Stop searching for a product after finding it.
 - continue: Skip damaged products during inspection and continue checking the remaining products.
-

9. What is Indentation in Python?

Answer:

Indentation means leaving spaces (usually 4 spaces) at the beginning of a line to define a block of code.

Real-Time Example:

In a classroom, all students of the same section sit together. Similarly, indented statements belong to the same block of code.

10. What is a Dictionary?

Answer:

A dictionary is a collection of key-value pairs. It stores data using keys to access values.

Real-Time Example:

A contact list in a mobile phone stores details like:

- Name → Bhavani
- Phone → 9876543210
- Email → bhavani@example.com

A Python dictionary is perfect for storing and retrieving this information using keys.

11. What is the difference between == and is?

Answer:

== and is are both comparison operators, but they check different things.

== checks equality → Do the two objects have the same value?

is checks identity → Are the two objects the same object in memory?

12. What is a Set? Why is it useful?

Answer:

A Set is an unordered collection of unique items in Python. The two key properties are: no duplicate values are allowed, and the elements have no index or order.

Real-Time Example

Suppose 500 students register for an event, but some register twice.

Using a Set ensures each student ID appears only once.

13. What is Exception Handling?

Answer:

Exception handling prevents a program from crashing when an unexpected error occurs. Instead of stopping the program, Python handles the error gracefully.

Real-Time Example

ATM Machine

If your account has insufficient balance, the ATM displays:

"Transaction Failed: Insufficient Balance."

It doesn't crash or stop working for everyone.

14. What is List Comprehension?

Answer:

List comprehension is a concise and efficient way to create lists in Python using a single line of code. It improves readability and often performs better than using a traditional loop.

15. What is Object-Oriented Programming (OOP)?

Answer:

Object-Oriented Programming (OOP) is a programming approach where we create classes and objects to model real-world entities. It helps make code reusable, organized, and easier to maintain.

Real-Time Example

Think of a Car.

Class: Car

Properties:

Color , Brand , Speed

Actions:

Start , Stop , Accelerate